

pr2

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1 Creuar data

Creuem les dades de totes les bases de dades, el codi el podem veure al annex¹

2 Analitzar exposició a fàrmacs i els riscos associats de fractura

2.1 Anàlisi descriptiva d'exposició en els grups de cas i control

Fem l'anàlisi ... ²

Table 1: Comparació de Medicaments i teràpies

	0	1	p	test
n	272	136		
insulinaDias (mean (SD))	30.13 (171.51)	65.08 (282.23)	0.122	
ActPrevantiulceros_IBP (%)			0.181	
0	127 (46.7)	56 (41.2)		
1	130 (47.8)	75 (55.1)		
2	15 (5.5)	4 (2.9)		
3	0 (0.0)	1 (0.7)		
ActualCortis_sistemics (%)			0.286	
0	265 (97.4)	133 (97.8)		
1	7 (2.6)	2 (1.5)		
2	0 (0.0)	1 (0.7)		
N06AA_SN = 1 (%)	6 (2.2)	4 (2.9)	0.910	
ADepreISRSCur = 1 (%)	51 (18.8)	34 (25.0)	0.182	
CortInh = 1 (%)	61 (22.4)	17 (12.5)	0.023	
PrevioCortis_sistemics (%)			0.530	
0	266 (97.8)	135 (99.3)		
1	5 (1.8)	1 (0.7)		
2	1 (0.4)	0 (0.0)		
N06AB_SN = 1 (%)	55 (20.2)	37 (27.2)	0.143	
N06A_SN = 1 (%)	74 (27.2)	51 (37.5)	0.044	
ADepreNoISRSDias (mean (SD))	69.70 (266.51)	112.32 (307.52)	0.149	
R03BA_SN = 1 (%)	22 (8.1)	2 (1.5)	0.014	

¹Annex A.1

²Annex A.2.1

	0	1	p	test
ADepreISRS6m = 1 (%)	39 (14.3)	27 (19.9)	0.199	
PrevioBisfosfonats = 3 (%)	1 (0.4)	0 (0.0)	1.000	
ActPrevCortis_sistemics (%)			0.477	
0	259 (95.2)	132 (97.1)		
1	12 (4.4)	3 (2.2)		
2	1 (0.4)	1 (0.7)		
ActualADO_glitazona (%)			0.286	
0	271 (99.6)	135 (99.3)		
1	0 (0.0)	1 (0.7)		
2	1 (0.4)	0 (0.0)		
ADepreISRS = 1 (%)	54 (19.9)	37 (27.2)	0.120	
ADepreISRSDias (mean (SD))	120.22 (400.11)	152.98 (388.67)	0.432	
N_SN = 1 (%)	197 (72.4)	104 (76.5)	0.450	
insulina = 1 (%)	14 (5.1)	10 (7.4)	0.503	
ADepreNoISRSCur = 1 (%)	36 (13.2)	30 (22.1)	0.032	
ActualCortis_inhalats (%)			0.430	
0	214 (78.7)	119 (87.5)		
1	20 (7.4)	8 (5.9)		
2	20 (7.4)	5 (3.7)		
3	11 (4.0)	3 (2.2)		
4	4 (1.5)	1 (0.7)		
5	2 (0.7)	0 (0.0)		
6	1 (0.4)	0 (0.0)		
H01_SN = 1 (%)	3 (1.1)	0 (0.0)	0.539	
N06AX_SN = 1 (%)	27 (9.9)	15 (11.0)	0.863	
ActPrevinsulina (%)			0.229	
0	258 (94.9)	126 (92.6)		
1	13 (4.8)	7 (5.1)		
2	0 (0.0)	2 (1.5)		
3	1 (0.4)	1 (0.7)		
Actualantiulceros_IBP (%)			0.282	
0	129 (47.4)	58 (42.6)		
1	130 (47.8)	73 (53.7)		
2	13 (4.8)	4 (2.9)		
3	0 (0.0)	1 (0.7)		
ActPrevCortis_inhalats (%)			0.312	
0	212 (77.9)	119 (87.5)		
1	20 (7.4)	8 (5.9)		
2	20 (7.4)	5 (3.7)		
3	11 (4.0)	3 (2.2)		
4	4 (1.5)	1 (0.7)		
5	2 (0.7)	0 (0.0)		
6	3 (1.1)	0 (0.0)		
BisfDias (mean (SD))	71.95 (273.97)	100.82 (360.65)	0.369	
ADO_no_glitazonaDias (mean (SD))	123.08 (443.98)	151.72 (456.99)	0.543	
ActPrevBisfosfonats (%)			0.275	
0	235 (86.4)	118 (86.8)		
1	30 (11.0)	18 (13.2)		
2	6 (2.2)	0 (0.0)		
3	1 (0.4)	0 (0.0)		
CortSistIniBf3m = 1 (%)	7 (2.6)	3 (2.2)	1.000	
CortInhIniBf3m = 1 (%)	46 (16.9)	12 (8.8)	0.040	

	0	1	p	test
Previoantidepressiu_no_ISRS (%)			0.066	
0	270 (99.3)	132 (97.1)		
1	1 (0.4)	4 (2.9)		
4	1 (0.4)	0 (0.0)		
ActPrevantidepressiu_ISRS (%)			0.170	
0	220 (80.9)	99 (72.8)		
1	47 (17.3)	34 (25.0)		
2	5 (1.8)	3 (2.2)		
CortSistDias (mean (SD))	18.85 (178.59)	10.36 (72.43)	0.595	
H02_SN = 1 (%)	15 (5.5)	6 (4.4)	0.812	
H_Dias (mean (SD))	76.87 (301.76)	88.26 (351.09)	0.734	
PrevioCortis_inhalats (%)			0.385	
0	266 (97.8)	136 (100.0)		
1	4 (1.5)	0 (0.0)		
3	1 (0.4)	0 (0.0)		
5	1 (0.4)	0 (0.0)		
CortSistExpLt3m = 1 (%)	5 (1.8)	3 (2.2)	1.000	
CortSistExpCat (%)			0.417	
0	267 (98.2)	133 (97.8)		
1	3 (1.1)	3 (2.2)		
2	2 (0.7)	0 (0.0)		
H_SN = 1 (%)	42 (15.4)	16 (11.8)	0.394	
Actualosteoporosi_no_bisfosfonat (%)			0.266	
0	208 (76.5)	114 (83.8)		
1	41 (15.1)	18 (13.2)		
2	11 (4.0)	2 (1.5)		
3	10 (3.7)	2 (1.5)		
4	2 (0.7)	0 (0.0)		
ADepreNoISRS6m = 1 (%)	23 (8.5)	19 (14.0)	0.120	
Bisf = 1 (%)	37 (13.6)	18 (13.2)	1.000	
PrevioADO_glitazona = 1 (%)	1 (0.4)	0 (0.0)	1.000	
CortInhExpCat (%)			0.080	
0	226 (83.1)	124 (91.2)		
1	34 (12.5)	8 (5.9)		
2	12 (4.4)	4 (2.9)		
Actualinsulina (%)			0.229	
0	258 (94.9)	126 (92.6)		
1	13 (4.8)	7 (5.1)		
2	0 (0.0)	2 (1.5)		
3	1 (0.4)	1 (0.7)		
CortInhExpLt3m = 1 (%)	46 (16.9)	12 (8.8)	0.040	
ActPrevADO_glitazona (%)			0.286	
0	271 (99.6)	135 (99.3)		
1	0 (0.0)	1 (0.7)		
3	1 (0.4)	0 (0.0)		
H02_Dias (mean (SD))	18.62 (178.54)	10.36 (72.43)	0.604	
CortSist = 1 (%)	13 (4.8)	6 (4.4)	1.000	
ADepreNoISRS = 1 (%)	36 (13.2)	31 (22.8)	0.021	
Previoinsulina = 0 (%)	272 (100.0)	136 (100.0)	NA	
R03BA_Dias (mean (SD))	28.75 (136.88)	5.72 (57.77)	0.061	
Actualantidepressiu_no_ISRS (%)			0.147	
0	236 (86.8)	107 (78.7)		

	0	1	p	test
1	27 (9.9)	18 (13.2)		
2	8 (2.9)	9 (6.6)		
3	1 (0.4)	1 (0.7)		
6	0 (0.0)	1 (0.7)		
ADO_no_glitazona = 1 (%)	43 (15.8)	27 (19.9)	0.378	
BisfExpCat (%)			0.317	
0	268 (98.5)	131 (96.3)		
1	1 (0.4)	2 (1.5)		
2	3 (1.1)	3 (2.2)		
Previoantiulceros_IBP = 1 (%)	4 (1.5)	2 (1.5)	1.000	
Actualantidepressiu_ISRS (%)			0.259	
0	221 (81.2)	101 (74.3)		
1	46 (16.9)	32 (23.5)		
2	5 (1.8)	3 (2.2)		
ADO_glitazonaDias (mean (SD))	0.04 (0.67)	1.74 (20.32)	0.168	
Previoantidepressiu_ISRS = 1 (%)	1 (0.4)	2 (1.5)	0.539	
N06_SN = 1 (%)	80 (29.4)	57 (41.9)	0.016	
ActualBisfosfonats (%)			0.185	
0	236 (86.8)	118 (86.8)		
1	30 (11.0)	18 (13.2)		
2	6 (2.2)	0 (0.0)		

2.2 Anàlisi logística “crua”

Fem l'anàlisi ... ³

Table 2: Anàlisi cru de les variables

	OR_Cru	IC_Inferior	IC_Superior	P_valor
ADO_glitazonaDias	1.02	0.99	NA	0.528
ADO_no_glitazona	1.32	0.77	2.240000e+00	0.308
ADO_no_glitazonaDias	1.00	1.00	1.000000e+00	0.544
Bisf	0.97	0.52	1.750000e+00	0.918
BisfDias	1.00	1.00	1.000000e+00	0.373
BisfExpCat	1.60	0.73	3.630000e+00	0.231
CortInh	0.49	0.27	8.700000e-01	0.018
CortSist	0.92	0.32	2.380000e+00	0.868
CortSistDias	1.00	1.00	1.000000e+00	0.605
CortSistIniBf3m	0.85	0.18	3.130000e+00	0.821
CortSistExpLt3m	1.20	0.24	4.980000e+00	0.801
CortSistExpCat	0.89	0.22	2.670000e+00	0.849
CortInhIniBf3m	0.48	0.23	9.000000e-01	0.030
CortInhExpLt3m	0.48	0.23	9.000000e-01	0.030
CortInhExpCat	0.62	0.36	9.900000e-01	0.061
ADepreISRS	1.51	0.93	2.440000e+00	0.094
ADepreISRSDias	1.00	1.00	1.000000e+00	0.433
ADepreNoISRS	1.94	1.13	3.300000e+00	0.015
ADepreNoISRSDias	1.00	1.00	1.000000e+00	0.157
insulina	1.46	0.61	3.360000e+00	0.374

³Annex A.2.2

	OR_Cru	IC_Inferior	IC_Superior	P_valor
insulinaDias	1.00	1.00	1.000000e+00	0.138
ADepreISRSCur	1.44	0.88	2.360000e+00	0.144
ADepreNoISRSCur	1.86	1.08	3.170000e+00	0.024
ADepreISRS6m	1.48	0.86	2.530000e+00	0.155
ADepreNoISRS6m	1.76	0.91	3.350000e+00	0.087
ActualBisfosfonats	0.86	0.49	1.460000e+00	0.594
Actualantidepressiu_no_ISRS	1.54	1.10	2.220000e+00	0.015
Actualosteoporosi_no_bisfosfonat	0.69	0.47	9.500000e-01	0.032
ActualCortis_inhalats	0.73	0.54	9.400000e-01	0.023
Actualantiulceros_IBP	1.14	0.80	1.620000e+00	0.472
ActualCortis_sistemics	1.12	0.32	3.460000e+00	0.843
Actualantidepressiu_ISRS	1.39	0.90	2.130000e+00	0.135
Actualinsulina	1.47	0.80	2.750000e+00	0.212
ActualADO_glitazona	1.00	0.07	6.760000e+00	1.000
PrevioCortis_sistemics	0.34	0.02	1.660000e+00	0.293
Previoantidepressiu_no_ISRS	1.22	0.45	3.390000e+00	0.649
PrevioCortis_inhalats	0.00	NA	6.710808e+14	0.978
Previoantiulceros_IBP	1.00	0.14	5.190000e+00	1.000
Previoantidepressiu_ISRS	4.04	0.38	8.749000e+01	0.256
PrevioADO_glitazona	0.00	NA	6.269489e+41	0.981
PrevioBisfosfonats	0.01	NA	8.558757e+13	0.981
ActPrevBisfosfonats	0.82	0.47	1.350000e+00	0.452
ActPrevCortis_inhalats	0.71	0.53	9.100000e-01	0.013
ActPrevantiulceros_IBP	1.13	0.80	1.610000e+00	0.476
ActPrevCortis_sistemics	0.75	0.25	1.820000e+00	0.550
ActPrevantidepressiu_ISRS	1.45	0.94	2.210000e+00	0.089
ActPrevinsulina	1.47	0.80	2.750000e+00	0.212
ActPrevADO_glitazona	0.85	0.07	3.280000e+00	0.824
H_SN	0.73	0.38	1.330000e+00	0.318
N_SN	1.24	0.77	2.010000e+00	0.382
H01_SN	0.00	NA	7.030794e+22	0.978
H02_SN	0.79	0.28	2.000000e+00	0.635
N06_SN	1.73	1.13	2.660000e+00	0.012
N06A_SN	1.61	1.03	2.490000e+00	0.034
N06AA_SN	1.34	0.34	4.780000e+00	0.652
N06AB_SN	1.47	0.91	2.380000e+00	0.113
N06AX_SN	1.12	0.56	2.170000e+00	0.730
R03BA_SN	0.17	0.03	5.900000e-01	0.017
H_Dias	1.00	1.00	1.000000e+00	0.734
H02_Dias	1.00	1.00	1.000000e+00	0.614
R03BA_Dias	1.00	0.99	1.000000e+00	0.103

2.3 Anàlisi logística ajustada per covariables

Fem l'anàlisi ... ⁴

⁴Annex A.2.3

Table 3: Anàlisi logística ajustada per covariables demogràfiques

	OR_Ajustat	IC_Inferior	IC_Superior	P_valor
ADO_glitazonaDias	1.02	0.99	NA	0.652
ADO_no_glitazona	1.14	0.42	2.940000e+00	0.786
ADO_no_glitazonaDias	1.00	1.00	1.000000e+00	0.225
Bisf	0.22	0.03	9.600000e-01	0.074
BisfDias	1.00	1.00	1.000000e+00	0.976
BisfExpCat	2965.53	0.00	NA	0.991
CortInh	0.26	0.05	9.300000e-01	0.059
CortSist	2.46	0.30	1.531000e+01	0.345
CortSistDias	1.00	0.98	1.020000e+00	0.649
CortSistIniBf3m	6.00	0.22	1.706600e+02	0.230
CortSistExpLt3m	6.00	0.22	1.706600e+02	0.230
CortSistExpCat	6.00	0.22	1.706600e+02	0.230
CortInhIniBf3m	0.14	0.01	7.300000e-01	0.047
CortInhExpLt3m	0.14	0.01	7.300000e-01	0.047
CortInhExpCat	0.11	0.01	5.100000e-01	0.018
ADepreISRS	0.69	0.22	2.000000e+00	0.509
ADepreISRSDias	1.00	1.00	1.000000e+00	0.468
ADepreNoISRS	3.80	1.21	1.230000e+01	0.022
ADepreNoISRSDias	1.00	1.00	1.000000e+00	0.190
insulina	1.15	0.08	8.470000e+00	0.900
insulinaDias	1.00	0.99	1.000000e+00	0.738
ADepreISRSCur	0.69	0.22	2.000000e+00	0.509
ADepreNoISRSCur	3.53	1.11	1.147000e+01	0.032
ADepreISRS6m	0.83	0.24	2.540000e+00	0.750
ADepreNoISRS6m	2.07	0.56	7.370000e+00	0.260
ActualBisfosfonats	0.20	0.03	7.800000e-01	0.047
Actualantidepressiu_no_ISRS	2.70	1.16	6.710000e+00	0.024
Actualosteoporosi_no_bisfosfonat	0.14	0.02	5.000000e-01	0.013
ActualCortis_inhalats	0.62	0.30	1.040000e+00	0.118
Actualantiulceros_IBP	1.87	0.90	4.000000e+00	0.096
ActualCortis_sistemics	4.85	0.88	4.435000e+01	0.096
Actualantidepressiu_ISRS	0.92	0.37	2.060000e+00	0.839
Actualinsulina	2.14	0.71	7.900000e+00	0.173
ActualADO_glitazona	1.57	0.08	1.130000e+01	0.666
PrevioCortis_sistemics	0.00	NA	4.348079e+63	0.989
Previoantidepressiu_no_ISRS	4944262.94	0.00	NA	0.988
PrevioCortis_inhalats	0.00	NA	4.319852e+53	0.989
Previoantiulceros_IBP	0.00	NA	1.939187e+63	0.989
PrevioADO_glitazona	0.00	NA	5.807498e+122	0.993
PrevioBisfosfonats	0.01	NA	8.343143e+40	0.993
ActPrevBisfosfonats	0.20	0.03	7.600000e-01	0.046
ActPrevCortis_inhalats	0.61	0.30	1.000000e+00	0.095
ActPrevantiulceros_IBP	1.73	0.84	3.650000e+00	0.136
ActPrevCortis_sistemics	2.76	0.59	1.385000e+01	0.178
ActPrevantidepressiu_ISRS	0.92	0.37	2.060000e+00	0.839
ActPrevinsulina	2.14	0.71	7.900000e+00	0.173
ActPrevADO_glitazona	1.19	0.08	4.870000e+00	0.829
H_SN	1.64	0.53	4.910000e+00	0.374
N_SN	0.64	0.23	1.840000e+00	0.393
H01_SN	0.00	NA	3.048635e+63	0.989

	OR_Ajustat	IC_Inferior	IC_Superior	P_valor
H02_SN	1.99	0.25	1.128000e+01	0.457
N06_SN	1.40	0.55	3.530000e+00	0.480
N06A_SN	1.18	0.44	3.070000e+00	0.740
N06AA_SN	1.14	0.05	1.054000e+01	0.917
N06AB_SN	0.69	0.22	2.000000e+00	0.509
N06AX_SN	2.24	0.53	9.050000e+00	0.255
R03BA_SN	0.08	0.00	9.900000e-01	0.150
H_Dias	1.00	1.00	1.000000e+00	0.340
H02_Dias	1.00	0.98	1.020000e+00	0.649
R03BA_Dias	0.99	NA	1.000000e+00	0.015

Observant la Taula 2 i la Taula 3

2.4 Estadístics descriptius, mesures de risc, inferències...

A Annex

A.1 Creuar data

```
library(haven)

uab_demo2 <- read_sas("uab_demo2.sas7bdat", NULL)
uab_drugs2 <- read_sas("uab_drugs2.sas7bdat", NULL)
uab_atcdrugs2 <- read_sas("uab_atc_drugs2.sas7bdat", NULL)

data1 <- merge(uab_demo2, uab_drugs2, by.x="PatNo", by.y="PatNo")
data <- merge(data1, uab_atcdrugs2, by.x="PatNo", by.y="PatNo")
```

A.2 Anàlisi d'exposició als fàrmacs i els riscos associats de fractura

A.2.1 Anàlisi descriptiva

```
library(dplyr)
library(tableone)

variables_farmacs <- c("insulinaDias", "ActPrevantiulceros_IBP", "ActualCortis_sistemics", "N06AA_SN",
  "ADepreISRSCur", "CortInh", "PrevioCortis_sistemics", "N06AB_SN", "N06A_SN",
  "ADepreNoISRSDias", "R03BA_SN", "ADepreISRS6m", "PrevioBisfosfonats",
  "ActPrevCortis_sistemics", "ActualADO_glitazona", "ADepreISRS", "ADepreISRSDias",
  "N_SN", "insulina", "ADepreNoISRSCur", "ActualCortis_inhalats", "H01_SN",
  "N06AX_SN", "ActPrevinsulina", "Actualantiulceros_IBP", "ActPrevCortis_inhalats",
  "BisfDias", "ADO_no_glitazonaDias", "ActPrevBisfosfonats", "CortSistIniBf3m",
  "CortInhIniBf3m", "Previoantidepressiu_no_ISRS", "ActPrevantidepressiu_ISRS",
  "CortSistDias", "H02_SN", "H_Dias", "PrevioCortis_inhalats", "CortSistExpLt3m",
  "CortSistExpCat", "H_SN", "Actualosteoporosi_no_bisfosfonat", "ADepreNoISRS6m",
  "Bisf", "PrevioADO_glitazona", "CortInhExpCat", "Actualinsulina", "CortInhExpLt3m",
  "ActPrevADO_glitazona", "H02_Dias", "CortSist", "ADepreNoISRS", "Previoinsulina",
  "R03BA_Dias", "Actualantidepressiu_no_ISRS", "ADO_no_glitazona", "BisfExpCat",
  "Previoantiulceros_IBP", "Actualantidepressiu_ISRS", "ADO_glitazonaDias",
  "Previoantidepressiu_ISRS", "N06_SN", "ActualBisfosfonats")

variables_cat=variables_farmacs[!grepl("Dias", variables_farmacs)]

taula_an <- CreateTableOne(
  vars = variables_farmacs ,
  strata = "Grup.x",
  data = data,
  factorVars = variables_cat
)

tbl=print(taula_an)
class(tbl)
```



```
library(knitr)
kable(tbl,caption = "Comparació de Medicaments i teràpies")
```

A.2.2 Anàlisi logística crua

```
variables_a_excloure <- c(
  "PatNo", "Grup", "Edat", "Sexe", "ServeiAlta", "PES",
  "TALLA", "IMC", "ALCOHOL", "Smoker", "Hipogonad", "Artritis_reumatoide",
  "Fractura", "NFractura", "FX_familia", "Diabetis", "tipus_diabetis",
  "Densitometries", "Osteoporosi", "neoplasia", "HiperTiroidisme",
  "Malnutricio", "Malabsorcio", "Malaltia_Hep_Cro", "OthRiskFract",
  "CountActualGE4",
  # per evitar errors
  "Grup.x", "Grup.y"
)

totes_les_columnes <- names(data)

variables_analisi <- setdiff(totes_les_columnes, variables_a_excloure)

resultats_2b <- lapply(variables_analisi, function(variable) {

  formula_model <- as.formula(paste("Grup ~", variable))

  model <- glm(formula_model, data = data, family = binomial)

  coefficients <- summary(model)$coefficients

  if(nrow(coefficients) > 1) {
    estimacio <- coef(model)[2]
    p_valor <- coefficients[2,4]

    OR <- exp(estimacio)
    IC <- exp(confint(model)[2, ])

    return(data.frame(
      OR_Cru = round(OR, 2),
      IC_Inferior = round(IC[1], 2),
      IC_Superior = round(IC[2], 2),
      P_valor = round(p_valor, 3)
    ))
  }
})

taula_final_2b <- do.call(rbind, resultats_2b)

tab <- print(taula_final_2b)

kable(tab, caption="\\label{tab:logcru}Anàlisi cru de les variables")
```

A.2.3 Anàlisi logística ajustada

```
covariables_demografiques <- c("Edat", "Sexe", "IMC", "ALCOHOL", "Smoker", "Osteporosi", "NFractura")
variables_farmac <- variables_analisi # ja definides amb setdiff(totes_les_columnes, variables_a_exclo
resultats_ajustats <- lapply(variables_farmac, function(variable) {

  formula_model <- as.formula(
    paste("Grup ~", variable, "+", paste(covariables_demografiques, collapse = " + "))
  )

  model <- glm(formula_model, data = data, family = binomial)
  coefficients <- summary(model)$coefficients

  if(variable %in% rownames(coefficients)) {
    estimacio <- coef(model)[variable]
    p_valor <- coefficients[variable,4]
    IC <- exp(confint(model)[variable, ])

    return(data.frame(
      OR_Ajustat = round(exp(estimacio), 2),
      IC_Inferior = round(IC[1], 2),
      IC_Superior = round(IC[2], 2),
      P_valor = round(p_valor, 3)
    ))
  }
})
taula_final_ajustada <- do.call(rbind, resultats_ajustats)
tab2 <- print(taula_final_ajustada)

kable(tab2, caption = "\\label{tab:logajust}Anàlisi logística ajustada per covariables demogràfiques")
```